

§1.7 — the two adjoints of $f^\#$: why

“existential” and “universal”

Freyd & Scedrov, §1.7 (opening definition)

The reader’s question: “in §1.7 the logos makes $f^\#$ a **left** adjoint? Isn’t it already a **right** adjoint in pre-logoi?” Answer: both, on opposite sides — the chain is $f \dashv f^\# \dashv f^{\#\#}$, and the two outer functors are the quantifiers $\exists$ and $\forall$. Lean: `Fredy/S1_70.lean` (`Logos`, `HasRightAdjointImage`, `DirectImage` = `existsAlong`).

§1. The picture: who passed what

$B = \text{Students} = \{\text{Ann}, \text{Bob}, \text{Cid}\}$, $C = \text{Courses} = \{\text{Alg}, \text{Top}\}$, and the predicate $P \subseteq B \times C$ records who passed what: $P(s, c) = \text{“student } s \text{ passed course } c\text{”}$.

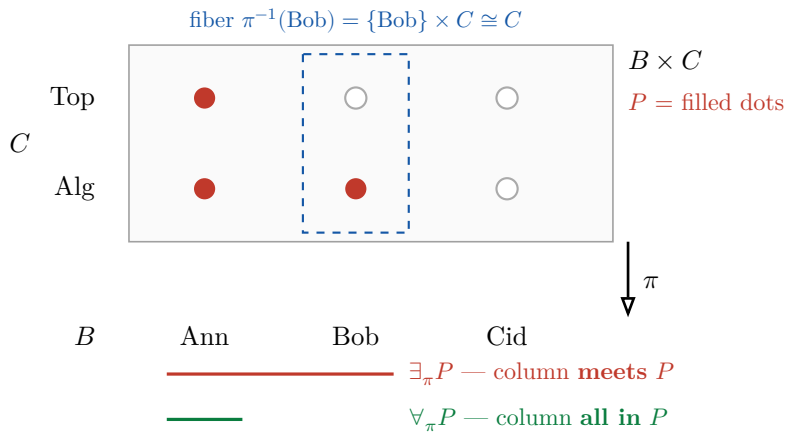


Figure 1: Each column is a fiber of π , a copy of C . Ann’s column is all filled $\rightarrow$ in $\forall$; Bob’s contains a filled dot $\rightarrow$ in $\exists$ only; Cid’s contains none $\rightarrow$ in neither.

Squashing P down along the projection $\pi : B \times C \rightarrow B$ gives two predicates on students:

$$\begin{aligned} \exists_\pi P &= \{s : \exists c. P(s, c)\} = \{\text{Ann}, \text{Bob}\} && // s \text{ passed some course} \\ \forall_\pi P &= \{s : \forall c. P(s, c)\} = \{\text{Ann}\} && // s \text{ passed every course} \end{aligned}$$

§2. Quantifiers are the adjoints of weakening

Three functions run between $\text{Sub}(B)$ and $\text{Sub}(B \times C)$. **Weakening** $\pi^\# : \text{Sub}(B) \rightarrow \text{Sub}(B \times C)$ sends $S' \subseteq B$ to $\{(s, c) : s \in S'\}$ // the students S' , paired with every course. $\exists_\pi$ and $\forall_\pi$ go the other way: input a whole predicate P (a set of pairs), output a predicate on students — the course variable c comes out **bound**. Binding a free variable is exactly what a quantifier does, and these two are the left and right adjoints of weakening:

$$\exists_\pi P \subseteq S' \text{ iff } P \subseteq \pi^\#(S'), \quad \pi^\#(S') \subseteq P \text{ iff } S' \subseteq \forall_\pi P.$$

Read the right-hand law on the picture: “every pair (s, c) with $s \in S'$ is a pass” iff “every student in S' passed every course” — the same sentence parsed two ways; that is all the adjunction says.

§3. Any map, not just projections: quantify over the fiber

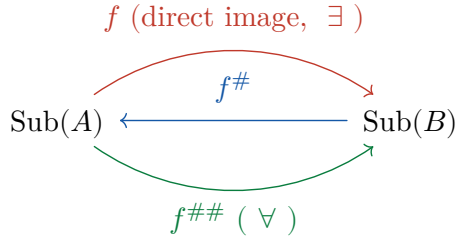
For an arbitrary map $f : A \rightarrow B$ in Set the same two adjoints of $f^\#$ exist; they quantify over the fibers $f^{-1}(s)$:

$$f(A') = \{s : \exists a \in f^{-1}(s). a \in A'\} \quad // \text{ the fiber meets } A'$$

$$f^{\#\#}(A') = \{s : f^{-1}(s) \subseteq A'\} \quad // \text{ the fiber is inside } A'$$

For $f = \pi$ the fiber over s is the column $\{s\} \times C$ — the **same** copy of C for every s — so “quantify over the fiber” becomes a bound variable c with fixed range, the literal first-order syntax $\exists c. P(s, c)$; the direct image π is $\exists_\pi$ and $\pi^{\#\#}$ is $\forall_\pi$.

§4. §1.7: a pre-logos has $\exists$, a logos adds $\forall$



In any regular category the direct image is already **left** adjoint to $f^\#$ (§1.51), so every pre-logos has $\exists$. A LOGOS (§1.7) requires each $\text{Sub}(A)$ to be a lattice and $f^\#$ to have a **right** adjoint $f^{\#\#}$ — it adds $\forall$. Not redundant: the old adjunction makes $f^\#$ a right adjoint (preserves intersections), the new one makes it also a left adjoint (preserves unions) — which is §1.711’s proof that a logos is a pre-logos.

Pre-logos = regular logic ($\exists$ only); logos = full first-order logic — and $f^{\#\#}$ along monics gives Heyting implication in each $\text{Sub}(A)$ (§1.712).